

Customer Churn Analysis

Executive Summary

An exploratory analysis of 7,043 telecom customer records (21 attributes covering demographics, account details, subscribed services, billing, and churn status) was conducted to identify the drivers of customer attrition. Overall, 26.5% of customers in the dataset have churned. Four factors stand out as the strongest predictors: contract length, tenure, use of technical/security add-on services, and payment method.

7,043 Total Customers <i>records analyzed</i>	26.5% Overall Churn Rate <i>1,869 customers</i>	42.7% Highest-Risk Segment <i>month-to-month contracts</i>	2.8% Lowest-Risk Segment <i>two-year contracts</i>
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Key Findings

Driver	Finding	Churn Rate
Contract Type	Churn drops sharply as contract commitment lengthens. Month-to-month customers are ~4x more likely to churn than one-year customers, and ~15x more likely than two-year customers.	42.7% / 11.3% / 2.8%
Tenure	Churn is concentrated among newer customers. Attrition is highest in the first few months of the relationship and declines steadily as tenure increases.	Skewed to low tenure
Senior Citizens	Senior citizens churn at nearly double the rate of non-seniors, despite making up a minority of the customer base.	41.7% vs 23.6%
Payment Method	Customers paying by electronic check churn at a much higher rate than those on mailed check, bank transfer, or credit card (all auto-pay or more “friction-free” methods).	45.3% vs 15–19%
Security / Support Add-ons	Customers without OnlineSecurity, OnlineBackup, DeviceProtection, or TechSupport churn noticeably more than subscribers; OnlineSecurity and TechSupport show the strongest retention effect.	Add-ons reduce churn
Streaming Services	StreamingTV and StreamingMovies show minimal correlation with churn — customers churn at similar rates whether or not they subscribe.	Limited impact
Gender	Male and female customers churn at essentially the same rate; gender is not a meaningful driver of attrition.	No material difference

Recommendations

- Incentivize longer-term contracts — targeted discounts or perks for month-to-month customers who switch to one- or two-year plans could meaningfully cut churn, given the sharp drop-off in attrition by contract length.
- Strengthen early-tenure onboarding — since churn is concentrated in the first months, a proactive onboarding or check-in program for new customers may reduce early attrition.
- Prioritize retention outreach to senior citizens — this segment churns at nearly double the average rate and would benefit from tailored support or offers.

- Investigate the electronic check experience — the payment method most associated with churn is also the most manual one; encouraging migration to autopay (bank transfer or credit card) may help, alongside understanding root causes (e.g., billing friction or financial hardship).
- Promote security and support add-ons — OnlineSecurity and TechSupport correlate with the strongest retention, making them good candidates for bundling or promotion, while streaming add-ons should not be relied on as a retention lever.

Methodology Note

Data was cleaned prior to analysis (blank TotalCharges values imputed as 0 and converted to numeric; SeniorCitizen recoded to a Yes/No categorical field). No duplicate customer records or missing values were found. Findings above are descriptive, based on cross-tabulations and visualizations of churn against each customer attribute; they highlight association, not confirmed causation. A predictive model (e.g., logistic regression) would be a natural next step to quantify each factor's independent effect and estimate churn risk at the individual customer level.